Research paper

A Chiral SU(2) Gauge-Sector Extension and the Quaia z~1-1.5 Angular-Mode Diagnostics

Technical regeneration with formulas, notation, data provenance, program inventory, and July 14-15, 2026 claim gates

Field	Value
Prepared for	Clive Boyd
Regenerated	July 15, 2026
Primary empirical focus	Quaia object-level redshift dipole diagnostics around z = 0.95-1.45
Theory focus	Phenomenological chiral SU(2)-like gauge-sector effective dark-energy component
Current claim status	Exploratory targeted follow-up; not a SU(2) detection and not a headline anisotropy claim

Claim boundary

The current evidence supports a disciplined follow-up program around a targeted Quaia redshift window. It does not establish a detected SU(2) angular symmetry, does not pass the all-window global look-elsewhere gate, and remains stressed by local catalog-quality and WISE-color controls.

Abstract

This paper formalizes a two-level research program. At the theoretical level, it describes how a chiral, SU(2)-like non-Abelian gauge sector could be parameterized as an effective late-time component in the Friedmann equation without claiming that ordinary Standard Model SU(2)_L is itself dark energy. At the empirical level, it documents the July 14-15, 2026 diagnostics on Quaia angular redshift modes, raw-MU supernova calibration, HHT residual screening, and SPARC rotation-curve posterior checks. The most interesting surviving empirical hook is a targeted Quaia mode in the z = 0.95-1.45 window: the high-stat targeted null gives joint p = 4.99975e-5 for the broad window, while external SFD dust, WISE tile-depth, Gaia stellar-density, and Quaia selection/depth controls do not absorb the mode. However, the global all-window look-elsewhere test remains ordinary, with p_global = 0.432713 for the strongest z1p00-1p50, |b|>35 row, and local catalog-quality/WISE-color strata produce large amplitude and direction changes. The correct conclusion is therefore not discovery, but a sharply defined follow-up gate: cross-catalog validation and stronger survey-template likelihoods must beat the catalog-quality explanation before the angular mode can be promoted.

Executive Claim Boundary

Gate	Observed status	Readout
Theoretical basis	SU(2) gauge sectors are mathematically serious and self-interacting.	Model-building support only; no dark-energy claim follows from Yang-Mills alone.
Targeted z window	broad 0.95-1.45 joint p = 4.99975e-5; narrow 1.15-1.35 joint p = 0.00444978.	Interesting targeted follow-up signal.
Global look-elsewhere	z1p00-1p50, |b|>35 p_global = 0.432713; priority p = 0.20076.	Fails headline anisotropy promotion.
Angular/footprint proxies	Amplitude ratios about 1.02; direction separations below 5 deg.	These simple angular and selection proxies do not explain the mode.
External maps	SFD, WISE depth, Gaia stellar density retain amplitude ratio about 0.98-1.10.	Dust/depth/stellar-density controls do not absorb the mode.
Gaia scan law	Scan-law preserving null joint p = 0.00349825; regression can reduce amplitude.	Rare under scan-preserving nulls, but still a material caveat.
Catalog quality	zerr, g, WISE colors, PM errors rotate directions by 50-104 deg or change amplitudes strongly.	Physical interpretation gate not passed.
Multipoles	Quadrupole-only readout stable; all-controls-plus-quadrupole has q/d mean = 19.1919.	Low-ell structure must be explained before a single-dipole story.
Adjacent streams	Raw-MU now calibration-budget first; SPARC is subset-win only; HHT is screening.	Contextual diagnostics, not independent SU(2) detections.

Acronyms and Variables

Acronym	Meaning
AIC	Akaike information criterion; lower values indicate better penalized fit.
BIC	Bayesian information criterion; in this project often used as a BIC-like score with random-effect priors.
CMB	Cosmic microwave background.
CPL	Chevallier-Polarski-Linder equation-of-state form w(a) = w0 + wa(1-a).
DES-Dovekie	Recalibrated Dark Energy Survey supernova release used in raw-MU diagnostics.
FITS	Flexible Image Transport System, used for Quaia catalog, randoms, and maps.
HHT	Hilbert-Huang transform, used here for residual-mode screening.
IMF	Intrinsic mode function in HHT decomposition.
IRSA	NASA/IPAC Infrared Science Archive.
LCDM	Lambda cold dark matter cosmological model.
PLAMB	Project shorthand for the logarithmic/optical-depth modified rotation or expansion diagnostics.
Quaia	Gaia-unWISE all-sky quasar catalog used for angular redshift-mode tests.
RAR	Radial acceleration relation baseline used in SPARC rotation-curve comparison.
SFD	Schlegel-Finkbeiner-Davis dust map.
SN	Type Ia supernova in this document.
SNR	Signal-to-noise ratio; for dipoles, sqrt(d^T Cov(d)^-1 d).
SPARC	Spitzer Photometry and Accurate Rotation Curves database.
SU(2)	Special unitary group of degree 2, the non-Abelian internal symmetry group considered here.
WISE	Wide-field Infrared Survey Explorer; unWISE/AllWISE information enters Quaia and depth controls.

Symbol	Meaning in this paper
z, zHEL, zCMB, zHD	Redshift, heliocentric redshift, CMB-frame redshift, and Hubble-diagram redshift with external-flow correction.
mu	Distance modulus, usually mu = 5 log10(D_L/Mpc) + 25.
H0	Present expansion rate in km/s/Mpc.
E(z)	Dimensionless expansion function H(z)/H0.
Omega_m, Omega_r, Omega_k, Omega_L	Matter, radiation, curvature, and cosmological-constant density fractions.
Omega_chi	Effective fractional density assigned to the proposed chiral SU(2)-like component.
w0_chi, wa_chi	CPL equation-of-state parameters for the effective chiral component.
A_mu^a	SU(2) gauge potential; mu is spacetime index, a is internal group index.

F_mu nu^a	Non-Abelian SU(2) field strength tensor.
g	Gauge coupling in field-theory formulas; also a Gaia G-band magnitude variable in catalog-quality tables.
Phi	Symmetry-breaking scalar/order parameter in the schematic field-theory extension.
lambda_chi	Coefficient of a parity-odd chiral term Phi F Ftilde.
n_i	Unit sky vector for source i in Galactic Cartesian coordinates.
d	Three-component fitted dipole vector in the Quaia redshift regression.
T_ki	Template/control value k evaluated at object i.
bcut	Minimum absolute Galactic latitude cut, |b| > bcut.

Historical and Theoretical Background

Yang-Mills local SU(2) symmetry

Yang and Mills introduced the modern local non-Abelian gauge principle: if an internal isotopic spin rotation is allowed to vary from point to point, an ordinary derivative no longer transforms covariantly. A gauge connection must be introduced, and its field strength contains a commutator/self-interaction term. This is the mathematical reason an SU(2)-like sector is a serious candidate for a dynamical cosmological component; it carries stress-energy and can self-interact. The same fact is also why the claim must be cautious: the existence of a non-Abelian gauge field does not itself imply a late-time dark-energy sector.

```
SU(2) = { U in C^(2x2) : U^dagger U = I, det(U) = 1 }
T^a = sigma^a / 2,    [T^a, T^b] = i epsilon^{abc} T^c
psi(x) -> S(x) psi(x),    D_mu = partial_mu - i g A_mu^a T^a
F_mu nu^a = partial_mu A_nu^a - partial_nu A_mu^a + g epsilon^{abc} A_mu^b A_nu^c
```

Chirality and electroweak precedent

The Standard Model already contains a chiral SU(2) structure: SU(2)_L x U(1)_Y breaks to U(1)_EM, with left-handed fermions transforming differently from right-handed fermions. This motivates a handedness-biased extension but does not identify ordinary weak-interaction vacuum energy with dark energy. The electroweak scale is far above the observed late-time dark-energy scale. The viable hypothesis is instead a residual, hidden, emergent, or effective SU(2)-like chiral sector whose parameters can be tested against cosmological data.

```
SU(2)_L x U(1)_Y -> U(1)_EM
```

Field-theory testbed

A schematic gravitationally coupled chiral gauge-sector action can be written as a Yang-Mills term, an order-parameter kinetic term, a potential, matter, and an optional parity-odd chiral term. The parity-odd term is a natural location for a handedness bias, but it must ultimately be constrained with parity-sensitive observables such as CMB TB/EB, cosmic birefringence, or chiral gravitational waves.

```
S = int d^4x sqrt(-g) [ R/(16 pi G) - (1/4) F_mu nu^a F_a^{mu nu}
    - (1/2) (D_mu Phi)^a (D^mu Phi)^a - V(Phi)
    + lambda_chi Phi F_mu nu^a Ftilde_a^{mu nu} + L_m ]
T_mu nu^(A) = F_mu alpha^a F_nu^{a alpha} - (1/4) g_mu nu F_alpha beta^a F^{a alpha beta}
```

Isotropic cosmic-triad ansatz

A single vector field normally selects a preferred spatial direction. An SU(2) field has a useful cosmological route: lock the internal group index to the spatial index. The cosmic-triad ansatz preserves isotropy at the background level while retaining non-Abelian dynamics.

```
A_i^a(t) = phi(t) delta_i^a,    A_0^a = 0
ds^2 = -dt^2 + a(t)^2 delta_ij dx^i dx^j
```

Phenomenological expansion-history branch

The first-stage project implementation should remain phenomenological. Rather than solving the full SU(2) field equations, the current fitting branch adds an effective density Omega_chi(z) to the Friedmann function and tests whether supernova, BAO, and CMB-distance data permit it under identical cuts and priors used for LCDM, KdS, FR, and PLAMB-style branches.

```
E^2(z) = Omega_m(1+z)^3 + Omega_r(1+z)^4 + Omega_k(1+z)^2 + Omega_L + Omega_chi(z)
w_chi(z) = w0_chi + wa_chi z/(1+z)
Omega_chi(z) = Omega_chi0 (1+z)^{3(1+w0_chi+wa_chi)} exp[-3 wa_chi z/(1+z)]
```

Statistical Methods

Quaia redshift dipole regression

The Quaia angular diagnostic fits the object-level Quaia redshift as a function of sky direction and optional standardized templates. For each redshift window and Galactic latitude cut, each object receives a unit vector n_i in Galactic Cartesian coordinates. The fitted vector d is interpreted only as an empirical angular mode, not as a physical velocity or field direction.

```
z_i = c0 + d . n_i + sum_k gamma_k standardize(T_{k,i}) + epsilon_i
A_dipole = ||d||
SNR_d = sqrt( d^T Cov(d)^(-1) d )
direction(d) = (l, b) from d / ||d||
```

Direction stability is summarized by pairwise angular separation across latitude cuts. A low separation means the fitted direction remains coherent when the mask changes. Template stability is summarized by the amplitude ratio relative to the no-template baseline and by rotation of the fitted direction.

Null p-values and look-elsewhere correction

```
p_upper = (1 + count(metric_mock >= metric_observed)) / (1 + N_mock)
p_lower = (1 + count(metric_mock <= metric_observed)) / (1 + N_mock)
p_global = probability that any scanned window/cut reaches the observed threshold
```

The targeted stability p-values use pre-selected windows after the direction-amplitude scan. They measure whether a selected follow-up target is unusual under nested random-sky mocks. The global look-elsewhere p-values scan a larger family and are the appropriate gate for a headline anisotropy claim. In this project, the targeted gate is interesting while the global gate remains ordinary.

Raw-MU hierarchical offset likelihood

```
r = MU_observed - MU_model(H0, beta)
r = X_dataset delta_dataset + X_survey delta_survey + noise
delta_dataset ~ Normal(0, sigma_dataset^2)
delta_survey ~ Normal(0, sigma_survey^2)
```

The July 14 supernova result promotes calibration-aware raw-MU offsets as the default diagnostic. This prevents an apparent exact-log or beta preference from being read as cosmology before dataset and survey calibration structure has been absorbed with explicit priors.

Information criteria

```
AIC = chi^2 + 2k
BIC = chi^2 + k ln(N)
Delta AIC = AIC_model - AIC_LCDM
```

Datasets Used

Dataset	Local/project artifact	Purpose
Quaia G<20 quasar catalog	Cosmology/quaia_G20.0.fits	Primary object-level redshift and sky-position source for angular-mode tests.
Quaia random catalog	Cosmology/quaia_G20p0_randoms_simple.fits	Random-sky and footprint comparison support.
Quaia selection map	.spyder-py3/selection_function_NSIDE64_G20.0.fits	Selection/depth proxy template.
SFD dust maps	external_datasets/sfd and sfd_dataverse_manifest_2026-07-15.json	External Galactic dust template gate.
Gaia nominal scan law	gaiascanlaw data/cache when available	Scan count and scan angle controls; scan-law preserving redshift shuffles.
AllWISE W1/W2 coverage	external_datasets/allwise_atlas_inventory/ allwise_p3am_cdd_w1w2_coverage.csv	Tile-depth control for WISE-related selection effects.
Gaia EDR3 density HiPS	external_datasets/gaia_dr3_density and gaia_edr3_density_hips	External stellar-density control.
Pantheon+SH0ES	PantheonPlusSH0ES/DataRelease; local raw-MU extracts	SN raw-MU calibration and redshift-frame audit.
DES-Dovekie	des-science/DES-SN5YR; local HD tables	SN raw-MU calibration and redshift-frame audit.
Union3.1 / UNITY	Union3p1_UNITY1p8 compressed rows	Compressed SN node included in hierarchical raw-MU joint likelihood.
SPARC rotation curves	SPARC local sample and point CSVs; filtered all-Q2 subsets	RAR/PLAMB posterior and stress tests.
HHT residual series	SN residual exports, BAO rows, chronometer rows	Residual-mode and surrogate screening.

Program Inventory

Program or script	Role in this paper
analyze_su2_quaia_locked_templates_multipoles_2026-07-15.py	Locked primary/control redshift windows; template and dipole-plus-quadrupole diagnostics.
analyze_su2_quaia_quality_strata_2026-07-15.py	Quartile splits by zerr, Gaia G/BP-RP, WISE W1/W2/color, and proper-motion error variables.
analyze_su2_quaia_quality_preserving_nulls_2026-07-15.py	Quality-stratified random-sky and quality-footprint redshift-shuffle nulls.
analyze_su2_quaia_footprint_shuffle_highstat_2026-07-15.py	High-stat 5,000 mock footprint-preserving redshift shuffles.
analyze_su2_quaia_external_dust_gate_2026-07-15.py	SFD dust, Quaia selection/depth, Gaia scan-law, and local catalog-quality template regressions.
analyze_su2_quaia_scanlaw_followup_2026-07-15.py	Scan-law preserving shuffles and residualization after scan-law controls.
analyze_su2_quaia_wise_stellar_gate_2026-07-15.py	AllWISE W1/W2 tile-depth and Gaia EDR3 stellar-density external template gates.

fetch_sfd_dataverse_2026-07-15.py	SFD dust acquisition and manifest capture.
fetch_gaia_dr3_density_hpx5_2026-07-15.py / fetch_gaia_dr3_density_uws_2026-07-15.py	Gaia density acquisition attempts and HiPS-backed fallback.
run_su2_quaia_overnight_highstat_pipeline.ps1	Overnight high-stat targeted stability and global look-elsewhere pipeline.
fit_sparc_hierarchical_map.py / analyze_sparc_posterior_run.py	SPARC MAP/posterior stress scoring and convergence readouts.
rawmu calibration-budget scripts	Hierarchical dataset/survey offset likelihood and zHD/zHEL/zCMB frame policy.
HHT residual export and resonance scripts	Historical residual screening, IMF summaries, and surrogate controls.

Research History Breakdown

Period/date	Milestone	Consequence for claim status
1954	Yang-Mills local isotopic gauge invariance establishes non-Abelian SU(2) self-interaction.	Mathematical foundation for gauge-sector models, not direct evidence for dark energy.
Electroweak era	SU(2)_L x U(1)_Y supplies a chiral Standard Model precedent.	Handedness is theoretically natural, but ordinary weak SU(2) is not the late-time component.
Pre-July 2026	Effective SU2_chiral_DE branch proposed with Omega_chi0, w0_chi, wa_chi, and dormant chi_bias.	Sets up a testable phenomenological branch.
July 14, 2026	Quaia direction-amplitude audit finds the stronger stable window at 0.95-1.45 and narrower target around 1.15-1.35.	Narrows the candidate angular target; broad symmetry wording rejected.
July 14, 2026	High-stat redshift-window mock catalogs generated before interpreting anisotropy scans.	Prevents premature interpretation of redshift-window scans.
July 14, 2026	Raw-MU likelihood promoted to hierarchical dataset/survey offsets; zHD preregistered as default frame.	Supernova exact-log preferences become calibration/frame diagnostics.
July 14, 2026	SPARC RAR rescue converges with max Rhat = 1.031.	Convergence issue fixed, but result remains subset-win only.
July 14, 2026	HHT/CMB-SN archives captured and summarized.	Useful screening method; not a confirmed coupling signal.
July 15, 2026	Overnight targeted and global Quaia mocks complete.	Targeted signal survives; global all-window look-elsewhere does not.
July 15, 2026	Locked template, quality strata, scan-law, WISE-depth, and stellar-density gates run.	External maps mostly do not absorb signal; local quality controls remain blocking caveat.

July 14-15 Quaia Results

Direction-amplitude perturbation

Window	Mean amplitude	Max SNR	Max separation	Readout
0.90-1.40	0.0023115	3.6887	17.33 deg	stable direction
0.95-1.45	0.0030202	4.7852	12.00 deg	strongest and stable
1.00-1.50	0.0023450	3.7059	17.94 deg	baseline stable
1.05-1.55	0.0013951	2.7582	61.34 deg	direction breaks
1.10-1.60	0.0010442	2.1157	76.01 deg	direction breaks

The scan does not support wording such as a stable broad SU(2) angular symmetry. It supports a targeted follow-up near z = 0.95-1.45, with a narrower 1.15-1.35 control window. Upward perturbations rotate the direction and weaken the amplitude.

Targeted stability vs global look-elsewhere

Target	Observed max SNR	Direction spread	p(SNR)	p(spread)	Joint p
broad 0.95-1.45	4.7852	13.4723 deg	0.000149993	0.0040998	0.0000499975
narrow 1.15-1.35	3.29694	18.5137 deg	0.0380481	0.0170991	0.00444978

Global look-elsewhere readout: all-window correction remains ordinary.

Target	bcut	Observed SNR	Global p	Priority p	z1p00-family p
z1p00-1p50	35	3.70594	0.432713	0.20076	0.0203959
z1p00-1p50	10	3.47955	0.70086	0.394521	0.0407918
z1p00-1p50	15	3.59778	0.563887	0.291542	0.0271946
z1p00-1p50	25	3.49857	0.677864	0.372326	0.0365927

Interpretation
The targeted p-values justify further work on a locked follow-up target. They do not override the global p-values. The global audit is the headline gate, and it is not passed.

Locked template and multipole controls

Model	Amp ratio mean	Amp ratio min	Max direction sep	SNR max	Readout
angular_systematics_proxy	1.0157	1.00692	4.51692 deg	4.83764	pass proxy gate
footprint_selection	1.02118	1.00522	2.4011 deg	4.88309	pass proxy gate
quadrupole_only	1.0126	1.00645	2.90399 deg	4.83856	dipole comparatively stable
catalog_quality	0.584129	0.295773	117.563 deg	4.78188	fails quality gate
all_controls	0.615275	0.335244	116.713 deg	4.74355	selection-sensitive
all_controls_plus_quadrupole	0.531886	0.308058	119.049 deg	4.10654	low-ell structure important

Multipole model	Amp ratio mean	q/d mean	q SNR max	Median delta BIC	Readout
quadrupole_only	1.0126	0.506064	4.45315	45.6913	dipole comparatively stable
all_controls_plus_quadrupole	0.531886	19.1919	9.67658	-102825	low-ell structure important

Quality strata and quality-preserving nulls

Variable	Q4/Q1 amp ratio	Max direction sep	Quality gate
zerr	1.73952	60.2607 deg	fail
g	2.29744	68.773 deg	fail
bp_rp	0.951058	103.846 deg	fail by rotation
w1	0.778735	53.2999 deg	fail
w2	1.40766	49.8262 deg	fail
w1_w2	0.775818	88.0069 deg	fail
pmra_error	2.35282	88.6125 deg	fail
pmdec_error	1.97269	67.1426 deg	fail

High-stat footprint-preserving redshift shuffle p-values for the locked primary window.

Quality variable	n_mocks	p_snr	p_pair_sep	p_coherence	Joint p
zerr	5000	0.00039992	0.0127974	0.00039992	0.00019996
g	5000	0.00059988	0.0117976	0.00059988	0.00039992
w1	5000	0.0263947	0.0219956	0.00959808	0.00379924
w2	5000	0.00159968	0.0163967	0.00119976	0.00079984
w1_w2	5000	0.009998	0.0585883	0.0079984	0.00339932
pmra_error	5000	0.00059988	0.0115977	0.00019996	0.00019996
pmdec_error	5000	0.00019996	0.0119976	0.00019996	0.00019996

The footprint-preserving shuffles are not nullifying the targeted signal, but the quartile splits show that the fitted mode is entangled with object-quality space. The most conservative reading is that the quality variables are not simple explanations by themselves, but they are strong enough to block physical promotion.

External dust, depth, scan-law, and stellar-density gates

Control group	Amp ratio	SNR max	Max direction sep	Readout
baseline_no_template	1.000000	4.7852	13.4723 deg	reference
sfd_dust_external	1.00644	4.66601	12.9543 deg	SFD dust does not absorb
quaia_selection_depth	1.03478	4.79898	13.5025 deg	selection/depth proxy does not absorb
sfd_plus_selection_depth	1.02617	4.71411	12.9733 deg	combined dust/depth stable
gaia_scanlaw_external	0.783971	3.36404	29.8825 deg	material scan-law caveat
sfd_selection_gaia_external	0.866391	3.60328	31.6722 deg	retains substantial residual
all_available_templates	0.499563	2.92202	48.2136 deg	too mixed; diagnostic only

Scan-law null	n_mocks	n_scan_bins	p_snr	p_pair	p_coherence	Joint p
scan-law preserving z shuffle	2000	63	0.00949525	0.0809595	0.0069965	0.00349825

Control group	Amp ratio	SNR max	Max direction sep	Readout
wise_depth_external	0.978169	4.75555	15.7795 deg	WISE depth does not absorb
gaia_stellar_density_external	1.09983	5.00263	16.0578 deg	stellar density does not absorb
wise_plus_stellar_external	1.06899	4.92309	17.965 deg	combined WISE/stellar stable
all_external_wise_stellar	0.920458	3.55534	35.1206 deg	substantial residual remains
all_external_plus_local_quality	0.711541	3.66543	36.7889 deg	local quality still matters

Adjacent Empirical Streams

Supernova raw-MU calibration and redshift frame

The July 14 raw-MU update changes the default supernova diagnostic. The correct default is now the hierarchical dataset+survey offset likelihood in the preregistered HD/zHD frame, with calibration-budget sensitivity shown explicitly. zHEL and zCMB are retained as controls rather than headline frames.

Reference result	Value
Reference row	HD, dataset+survey, 25 mmag survey / 50 mmag dataset, FR_BETA05_H0fixed675
N	3422
Calibration random effects	27
Data chi2 at MAP	3092.819
Calibration prior chi2	19.154
BIC-like score	3146.032
Beta-free H0-fixed beta	0.518263
No-offset HD beta-free control	beta = 0.466532, BIC-like = 4118.004

SPARC PLAMB/RAR status

The RAR convergence-rescue run achieved the pre-run target and changed some stress readouts, but the SPARC result remains a subset-win story, not a full-sample PLAMB win.

Metric	Value or readout
RAR rescue convergence	5 chains, 3000 saved per chain, max Rhat = 1.031
All-Q2 baseline delta log score	+28.941; 65 better / 92 worse galaxies
High-quality Q1 subset	+379.463; strong subset win
Low-acceleration outer points	+48.508; reversed to positive
High inclination	-358.865; major failure
Gas dominated	-28.175; failure
Claim boundary	RAR convergence rescued; PLAMB remains subset-win only.

HHT CMB/SN residual screening

The HHT archive is valuable as a residual-screening workflow. It should not be presented as a confirmed CMB-SN coupling signal until a new preregistered run separates probes, controls survey/window effects, and passes global/surrogate correction.

HHT readout	Value
SN residual weighted RMS	LCDM 0.15159; SU2 0.15548; SU2R 0.15542
Extended SU2-minus-LCDM IMF energy fractions	IMF1 0.6177; IMF2 0.4290; IMF3 0.2068
Surrogate control	max-energy upper-tail p = 0.9703; sum-energy upper-tail p = 0.9604
Older IMF2-vs-BAO baseline	r_DM = 0.1006, p = 0.0078; r_DH = -0.1037, p = 0.0061
Claim boundary	Candidate-feature and method-development evidence only.

Interpretation and Next Tests

The current SU(2)/Quaia state is unusually productive because the result is neither dead nor promotable. Dust, WISE tile depth, Gaia stellar density, and the local selection map do not simply erase the targeted z = 0.95-1.45 mode. At the same time, all-window look-elsewhere correction and local quality/WISE-color entanglement prevent discovery language.

Next test	Reason	Promotion criterion
Cross-catalog angular validation	Quaia quality variables may encode catalog construction or photometric-redshift behavior.	Same z-window direction and amplitude class appears in an independent quasar/LSS catalog with a preregistered mask.
External template likelihood	Current regressions are linear template gates, not a full generative selection model.	Mode survives SFD, WISE depth, Gaia density, scan-law, and survey-depth likelihood with amplitude ratio >0.8 and direction rotation <30 deg.
Quality-color causal audit	w1, w2, w1_w2, zerr, G, and proper-motion errors are the strongest caveats.	Mode remains after matched quality/color reweighting or after physically motivated quality cuts fixed before outcome inspection.
Global null with fixed locked target	The all-window global p-value is ordinary.	A future pre-registered target family gives a globally corrected p small enough for headline language.
Multipole decomposition	All-controls-plus-quadrupole shows low-ell structure.	Coherent dipole/quadrupole pattern is explained by a physical or survey model, not just a marginal dipole.
Theory-to-data bridge	Expansion history alone cannot establish chirality.	A predicted angular/parity observable is derived from the SU(2) sector and beats survey-template alternatives.

Bottom-line conclusion

Use the phrase: targeted Quaia z~1-1.5 angular-mode follow-up. Do not use: detected SU(2) angular symmetry, cosmological anisotropy detection, or full proof of a chiral gauge-sector dark-energy component.

Reproducibility and Archived Artifacts

Artifact group	Representative files
July 15 SU2/Quaia results	github_export/results/2026-07-15/su2_quaia/ and outputs/su2_quaia_*_2026-07-15.*
July 14 SU2/Quaia baseline	github_export/results/2026-07-14/su2_quaia/ and outputs/su2_quaia_*_2026-07-14.*
Code capture	github_export/code/quaia_su2/ plus copied scripts in outputs/
Raw-MU calibration	github_export/results/2026-07-14/rawmu/ and rawmu_calibration_budget_* files
SPARC posterior	github_export/results/2026-07-14/sparc/ and sparc_filtered_rar_rescue_readout_2026-07-14.md
HHT CMB/SN archive	github_export/results/2026-07-14/hht_cmb_sn/ and hht_cmb_sn_scan_readout_2026-07-14.md
Research papers	github_export/docs/papers/ for citation-facing PDFs; local editable DOCX stored in outputs/

References and Source Links

- Yang, C. N., and Mills, R. L. (1954). Conservation of Isotopic Spin and Isotopic Gauge Invariance. Physical Review 96, 191-195.
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- AllWISE and IRSA programmatic access documentation: https://irsa.ipac.caltech.edu/ibe/ibe_search_api.html and IRSA AllWISE documentation pages.
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- gaiascanlaw package and source: https://pypi.org/project/gaiascanlaw/ and https://github.com/zpenoyre/gaiascanlaw.
- Pantheon+SH0ES data release: https://github.com/PantheonPlusSH0ES/DataRelease.
- DES-Dovekie / DES-SN5YR repository: https://github.com/des-science/DES-SN5YR.
- Union3 / UNITY1.5 reference: Rubin et al., Union Through UNITY, arXiv:2311.12098.
- Project generated reports listed in the reproducibility table, especially su2_quaia_overnight_highstat_status_2026-07-15.md, su2_quaia_locked_template_multipole_report_2026-07-15.md, su2_quaia_quality_strata_readout_2026-07-15.md, su2_quaia_external_dust_gate_readout_2026-07-15.md, su2_quaia_scanlaw_followup_readout_2026-07-15.md, and su2_quaia_wise_stellar_gate_readout_2026-07-15.md.

Appendix: Acceptance Gates

Gate	Pass threshold used in practice	Current status
Targeted stability	Low targeted p-values for SNR and direction coherence.	Passed for follow-up only.
Global look-elsewhere	Small all-window global p-value.	Not passed; p_global = 0.432713 for key row.
Template stability	Amplitude ratio not substantially suppressed; direction rotation below about 30 deg.	Passed for SFD, WISE depth, Gaia density; partial for scan law.
Quality robustness	Quartile/matched-quality tests do not rotate or amplify/suppress the mode materially.	Not passed.
Multipole coherence	Low-ell pattern supports a coherent physical model.	Open; all-controls quadrupole suggests low-ell complexity.
Adjacent diagnostics	SN, HHT, and SPARC pass their own preregistered gates.	Mixed; raw-MU policy improved, SPARC/HHT remain bounded.
Theory prediction	SU(2) field model predicts the observed angular/parity signal quantitatively.	Not yet implemented.